GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 09



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Number System

Topics to be Covered



Topic-1 More on Numbers

Topic-2 Counting Theory

[MCQ]

#Q. Find the number of zeroes in following multiplication:

Assignment

5 x 10 x 15 x 20 x 25..... x 50 ?

$= 5\times1\times5\times2\times5\times3\times5\times4\times\ldots\ldots\times5\times10$

$= 5^{10}\times(10!)$

$\frac{10}{5}\Rightarrow 2$

$\frac{2}{5}\Rightarrow\frac{0}{2}$

$\frac{10}{2}\Rightarrow 5$

$\frac{5}{2}\Rightarrow 2$

$\frac{2}{2}\Rightarrow 1$

$\frac{1}{2}\Rightarrow 0$

5 → 10+2 = 12

2 → 8

A 48

B 12

C 24

D 8

[MCQ]

#Q. How many squares are there in a chessboard?

Assignment

Size

1×1 ⇒

2×2 ⇒

3×3 ⇒

4×4 ⇒

5×5 ⇒

6×6 ⇒

7×7 ⇒

8×8 ⇒

204

No.

8×8 = 64

7×7 = 49

6×6 = 36

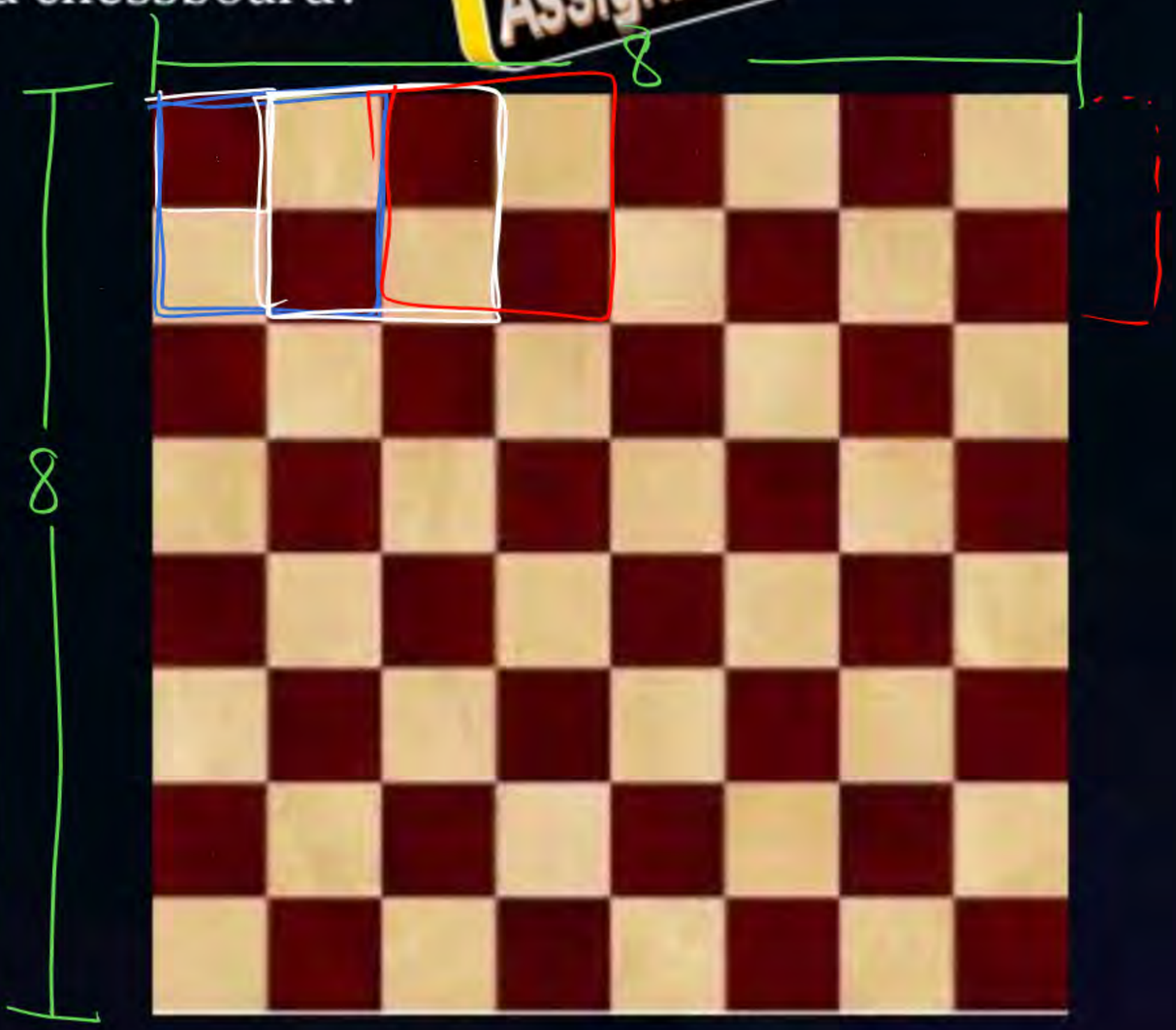
5×5 = 25

4×4 = 16

3×3 = 9

2×2 = 4

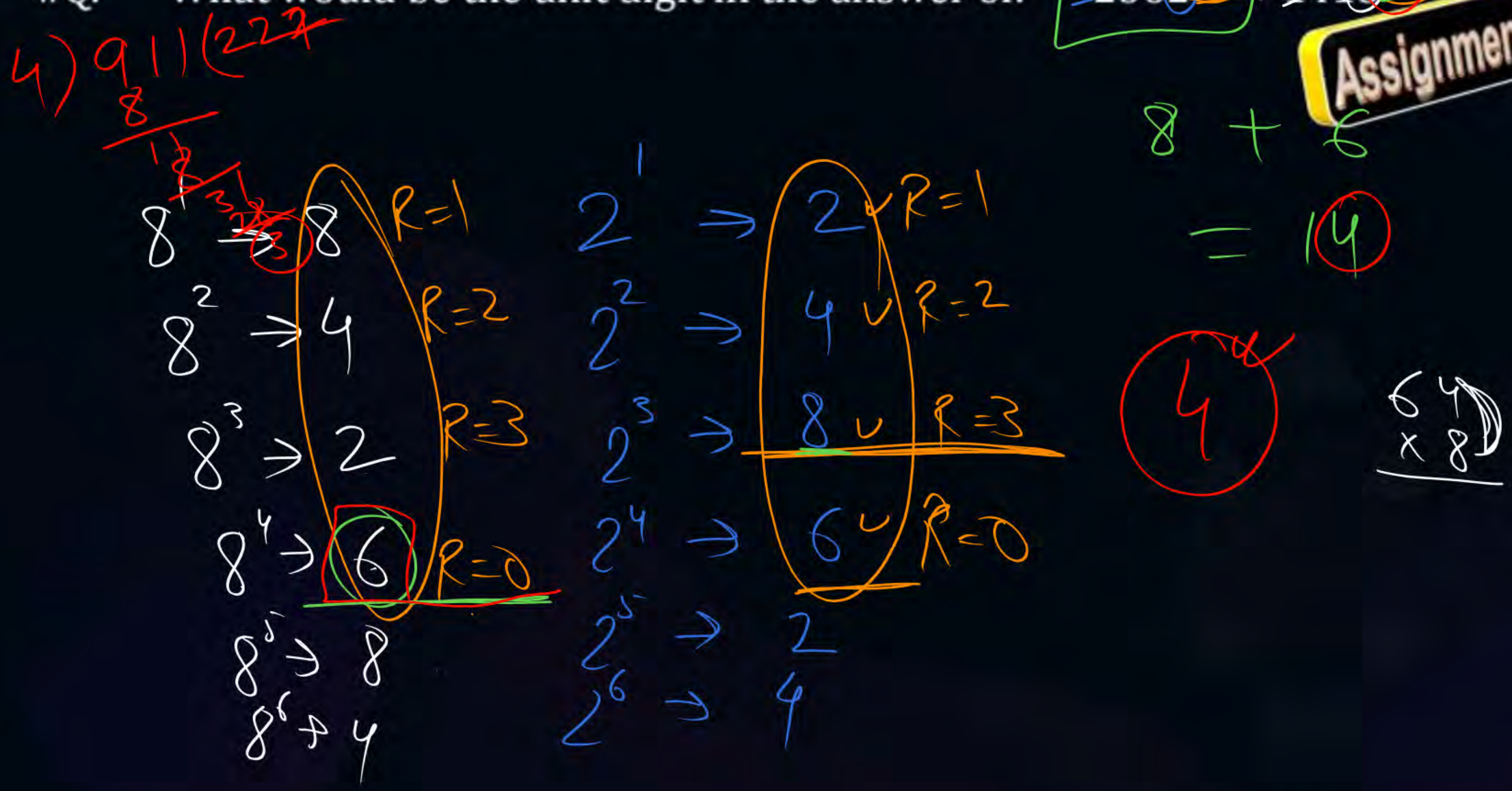
1×1 = 1



[MCQ]

#Q. What would be the unit digit in the answer of: $2362^{911} + 1410^{216}$

Assignment



[MCQ]

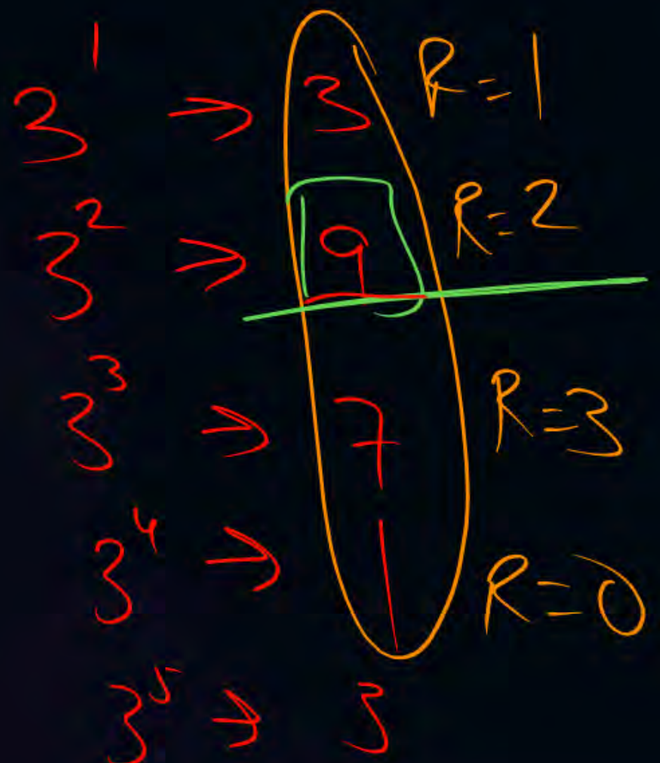
#Q. What would be the unit digit in the answer of: $2913^{1902} + 1647^{460}$?

A 0

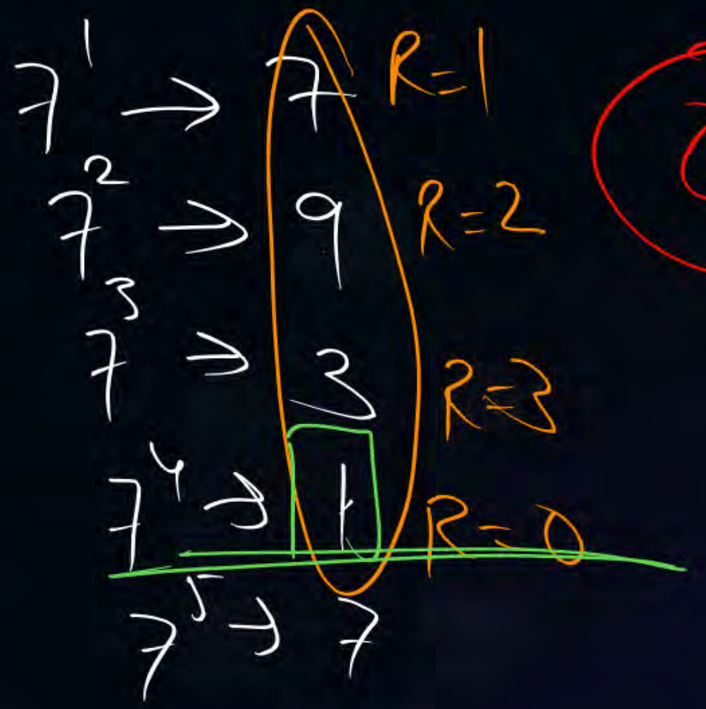
B 2

C 3

D 5



9 + 1 = 10



[MCQ] Base last digit 2, 3, 7, 8

#Q. What would be the unit digit in the answer of: $9326^{397} + 1475^{363}$

6 + 5

= 11

Base last digit

0, 1, 5, 6

Same No.

Base last digit

odd 4 or 9

odd → 4

even → 6

odd → 9

even → 1

$6^1 \Rightarrow 6$

$6^2 \Rightarrow 6$

$6^3 \Rightarrow 6$

$5^1 \Rightarrow 5$

$5^2 \Rightarrow 5$

$5^3 \Rightarrow 5$

$5^4 \Rightarrow 5$

1 → 4 → 4 R=1

2 → 4 → 6 R=0

3 → 4 → 4

[MCQ]

#Q. What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1 ?

A 9997

B 9793

C 9895

D 9487

2 | 6, 9, 12, 17

3 | 3, 9, 6, 17

1, 3, 2, 17

$2 \times 3 \times 3 \times 2 \times 17$

$= 612 + 1 = 613$

36×17

612

1224

1836

9792 (612×16)

(612×17)

$9792 + 1$

$= 9793$

Divisibility Rules:

6+8+9+4+7+8

68947|8

- 2: Given No. unit digit is 0 OR Multiple of 2.

- 3: → Sum of the digits of a given no. should be divisible by 3

- 4: (last two digit 00 OR Multiple of 4.

4 → 2 (2)

8 → 2 (3)

→ last three digits of given no 000 OR Multiple of 8

Divisibility Rules:

25 ⇒ 5² ②

↳ last two digits 00 OR multiple of 25

- 5: ⇒ last digit of given no should be 0 OR Multiple of 5

- 6: ⇒ 6 = 2 × 3 (e e)

- 7: ?

489318

7^9

48934 49

48934 ~ 18

= 48916

46 ~ 18

= 28

2 ~ 16 = 14

4891 ~ 12

= 4879

487 ~ 18 = 469

1 ~ 8

= 7

Twin Prime & Co-Prime Numbers:

(3, 5)

(13, 11)

(5, 7)

Any two Prime No with a difference of two

Co-prime

9 → 1, 3, 9

32 → 1, 2, 4, 8, 16, 32

* Any two consecutive No.
* Any two distinct Prime No.

Divisibility Rules:

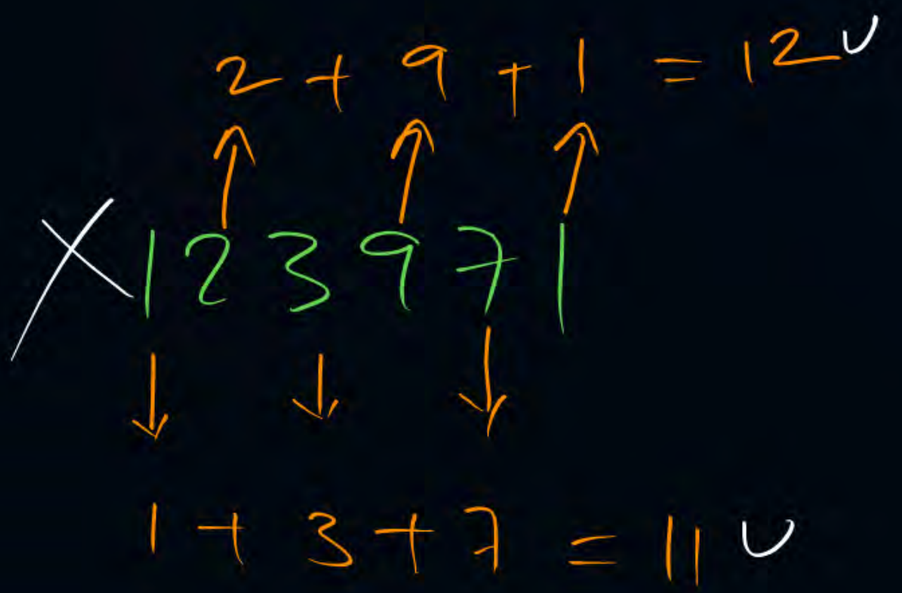
- 8: → $2^{(3)}$

  Last three digits of a given no 000 OR Multiple of 8

- 9: Sum of digits should be divisible of 9

- 10: last digit of a given no should be 0

Divisibility Rules:



- 11:
- 12: $= 3 \times 4$

2×7

- 14:

Divisibility Rules:

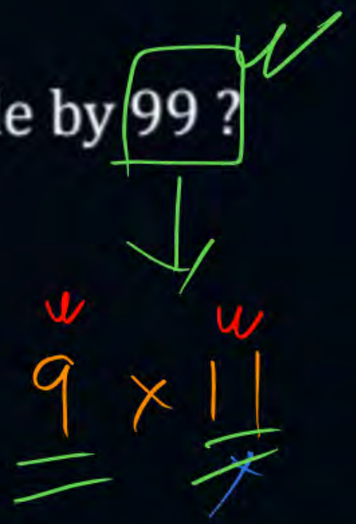
- 15: ⇒ 3×5
- 16: ⇒ 2^{4} last four digits of given no. 0000 OR Multiple of 16
- 18: ⇒ 2×9

Divisibility Rules:

- 20: → 4×5
- 40: → 8×5
- 80: → 5×16

[MCQ]

#Q. Which largest number of 5 digits is divisible by 99 ?



A. 99999

B. 99981

C. 99909

D. 99990

[MCQ]

Assignment

#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

A 216

B 1728

C 512

D 360

[MCQ]

Assignment

#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

A 13

B 14

C 17

D 16

[MCQ]

Assignment

#Q. Find the largest four digit number that is exactly divisible by 88.

A 9844

B 9768

C 8894

D 9944

Numbers

WORK

Counting

OR या (+)

AND और (×)

Counting Theory

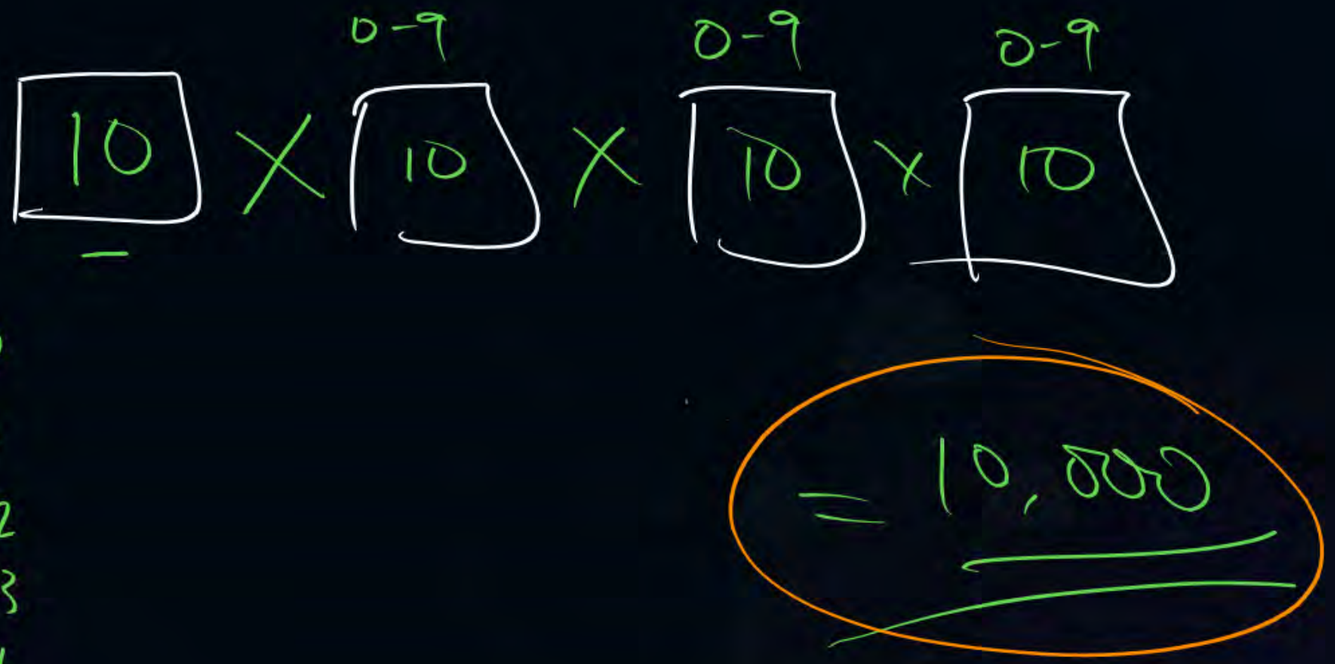
ATM

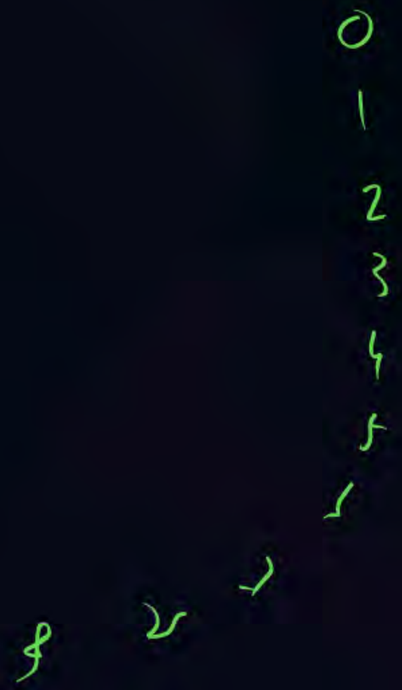
New PIN

4 digits

[MCQ]

#Q. In how many different ways you can choose your ATM PIN ?





[MCQ]

#Q. In how many different ways you can select a six digit phone password/passcode ?

0-9 0-9

10 × 10 × 10 × 10 × 10 × 10

= 1,000,000

[MCQ]

#Q. How many four digits number can be formed using 1, 2, 3, 4 & 5?

Rep ✓

Th	H	Te	U
5	× 5	× 5	× 5

= 625

Rep ✗ NOT ALLOWED

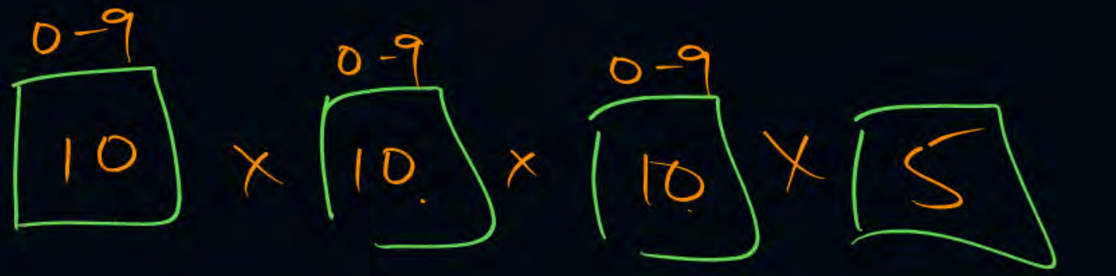
Th ✓	H ✓	Th	U
5	× 4	× 3	× 2

(5-1)(5-2)(5-3)

= 120

[MCQ]

#Q. In how many different ways you can choose your ATM PIN which is an even ?



$= 5000$

0
2
4
6
8

[MCQ]

#Q. How many three digits number can be formed using digits 0 to 9?

Rep ✓

H	Ten	U
9	10	10

$\not\emptyset$ (10-1)

$= 900$

Rep NOT ALLOWED

H ✓	Ten ✓	U
9	9	8

$\not\emptyset$ (10-1) (10-1) (10-2)

$= 81 \times 8 = 648$

[MCQ]

#Q. In how many ways four letters can be posted in 6 post boxes, if each box can take any number letters?

L_1 L_2 L_3 L_4

$6 \times 6 \times 6 \times 6$

$= 36 \times 36 = 1296$

A 4096

B 1500

C 24

D 1296

[MCQ]

Assignment

#Q. In how many ways can you arrange 5 Novels on a shelf?

A 720

B 120

C 360

D 300

[MCQ]

PYQ

#Q. Four persons A, B, C and D are to be seated in a row. C should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is?

Assignment

A 6

B 9

C 18

D 24

[MCQ]

PYQ

#Q. Five different book (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is?

Assignment

A 12

B 120

C 6

D 2

[MCQ]

Assignment

#Q. A license plate begins with 3 letters. If the possible letters are A, B, C, D and E, how many different ways these letters can be written if no letter is used more than once?

A 720

B 120

C 60

D 30

[MCQ]

Assignment

#Q. In how many different ways can the letters of the word 'PLEADING' be arranged in such a way that the vowels always come together?

A 720

B 4320

C 5040

D 3600



2 mins Summary

Topic Counting Theory

THANK - YOU